Britani Porwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
Shri Kasera Bazar Vidya Niketan	Class XII (CBSE)	2019	72.40%
Chameli Devi Public School	Class X (CBSE)	2017	10.00%

PROJECTS

SimLingo: Vision-Only Closed-Loop Autonomous Navigation (SOTA)

- End-to-end deployed and evaluated SimLingo (winning entry at CARLA leaderboard), a camera-only Vision-Language-Action (VLA) model for closed-loop physical execution without 3D LiDAR hardware dependencies.
- Configured LoRA fine-tuning on an InternVL2-1B backbone across multi-task objectives: waypoint control, driving-scene VQA (DriveLM), and language-action instruction alignment (Action Dreaming).
- Investigated disentangled control heads (decoupling 2D path geometry from speed profiles), experimentally proving that separating lateral steering from longitudinal dynamics mitigates autoregressive text hallucinations and eliminates static collisions.
- Automated evaluation on the Bench2Drive leaderboard in CARLA, using Driving Score (DS), Route Completion (RC), and inference latency as the metrics under simulated production constraints.

THESIS

Research Interest: Scalable Vision-Language-Action Models for Long-Horizon Embodied AI

- Embodied Perception & Navigation: Conducting research on multimodal foundation models for Vision-Language Navigation (VLN), enabling embodied agents to execute natural language instructions and long-horizon decision-making in continuous 3D environments.
- Data-Efficient Reasoning Paradigms: Formulating auxiliary Chain-of-Thought (CoT) supervision, instruction reconstruction, and receding-horizon action planning techniques to improve zero-shot generalization while minimizing explicit test-time reasoning overhead.
- Simulation Benchmarking & Evaluation: Benchmarking transformer-based navigation architectures on Habitat-Lab across Matterport3D, R2R, RxR, and VLN-CE datasets, quantifying navigation success rate, planning efficiency, and trajectory fidelity in unseen environments.

WORK EXPERIENCE

Solutions Engineer

Deqode Solutions × Dell Technologies

- AI Search Model Production Deployment: Integrated Coveo AI modules (Smart Snippets, Content Recommendation, In-Product Experience) into Dell's global search page, boosting customer conversion rates by 37%.
- Multilingual Data Pipeline: Built dynamic translation and localization pipelines across AEM architectures, supporting 76 global locales for high-throughput enterprise portals.

Software Engineer - Web3 & Distributed Applications

Deqode Solutions

- Blockchain Web Applications: Architected and deployed responsive, user-centric Web3 interfaces using React, Next.js, and TypeScript, integrating decentralized wallet hooks (Ethers.js, Wagmi, RainbowKit) for seamless on-chain transactions.
- Smart Contract Integration: Engineered secure frontend modules for real-time auctions, NFT bidding, reward coins, and multi-wallet authentication, directly interfacing with Ethereum smart contracts via Alchemy APIs.
- Performance Optimization: Optimized rendering pipelines and state synchronization for heavy visual assets and real-time activity feeds, significantly reducing UI latency and build admin access control panel entirely from scratch.

- Tech Stack: Next.js, React.js, TypeScript, Ethers.js, Web3.js, Node.js, Express.js, MongoDB, REST APIs.

SKILLS

Deep Learning : PyTorch, TensorFlow, Hugging Face Transformers, LoRA, OpenCV

Core ML: Scikit-Learn, XGBoost, NumPy, SciPy, Pandas

Languages : Python ,C++ ,SQL ,JavaScript ,TypeScript

Tools : CUDA, Git, Docker, Postman, Jira, Jupyter Notebook, Google Colab, Kaggle, VS Code, Linux

ACHIEVEMENTS & LEADERSHIP

Vigyan Vidushi Scholar (TIFR STCS 2026): Selected among the top 30 female researchers nationwide for the prestigious, fully-funded 2-week theoretical computer science & machine learning residential research program at Tata Institute of Fundamental Research (TIFR), Mumbai.

Assistant Manager (PR & Media Relations), E-Cell IITH: Coordinated E-Merge (IITH's annual entrepreneurship summit), driving outreach campaigns, media partnerships, and speaker relations with industry leaders.

Hostel Representative, Maitreyi Hall of Residence (IITH): Elected student head overseeing operations for Maiteryi Hostel block; acting as the primary moderator between students and hostel authorities and managed logistics for inter hostel cultural and sports events during annual fest.